Session – 5

Task-1.

List 3 examples of 'population' and 'sample' in the context of apps you use daily (like Instagram followers, Zomato restaurants, or Flipkart products). For each, clearly define what would be considered the population and what would be a sample.

Solution:

App / Context	Population	Sample
Instagram Followers	**All followers of an Instagram account**	**500 randomly selected followers surveyed about their content preferences**
Zomato Restaurants	**All restaurants listed on Zomato in India**	**1,000 randomly selected restaurants analyzed for ratings and prices**
Flipkart Products	**All products available on Flipkart**	**2,000 randomly selected products analyzed for ratings and prices**

Population = the complete group we want to study.
Sample = a smaller group selected from the population for analysis.

Task-2.

Write a Python script that simulates random sampling: given a list of 1000 fake user IDs (e.g., user1 to user1000), select 50 random users to simulate a survey sample. Print the selected user IDs.

Hint: Use the random.sample() function from Python's random module.

Solution:

```
import random


# Create 1000 fake user IDs

users = [f"user{i}" for i in range(1, 1001)]


# Select 50 random users

sample = random.sample(users, 50)


# Print selected users

print("Selected 50 users:")
```

```
for user in sample:
    print(user)
```

Output:

Selected 50 users:

user47

user812

user193

user621

user5

...

Task-3.

Imagine you are analyzing food delivery times on Swiggy. Explain in 2-3 sentences why you would use sampling instead of collecting data from every single order ever made.

Solution:

Collecting and analyzing every Swiggy order ever made would require a huge amount of data, time, storage, and computational resources. A properly selected sample can provide a reliable estimate of overall delivery times much more efficiently while reducing the cost and complexity of analysis.

Task-4.

Given this scenario: A survey about music preferences is only sent to users who recently streamed Bollywood songs on Spotify. Identify and explain the type of bias this introduces in the sample.

Solution:

The survey introduces **selection bias (sampling bias)** because it is only sent to users who recently streamed Bollywood songs.

These users are more likely to prefer or listen to Bollywood music, so they do not represent the entire Spotify user population. As a result, the survey may **overestimate the popularity of Bollywood music**.

Task-5.

Choose any two types of sampling (for example: random sampling, stratified sampling, systematic sampling, cluster sampling). For each, describe how you would use it to select a sample of users from BookMyShow to study movie booking habits.

Solution:

A. Random Sampling

Create a list of BookMyShow users and randomly select, for example, **1,000 users** from the entire user base.

This gives every user an equal chance of being selected and can provide a representative sample of overall movie-booking habits.

B. Stratified Sampling

Divide BookMyShow users into groups (**strata**) based on characteristics such as **age groups**: 18–25, 26–35, 36–50, and 50+.

Then randomly select users from each age group, ensuring that all important age groups are represented in the study.